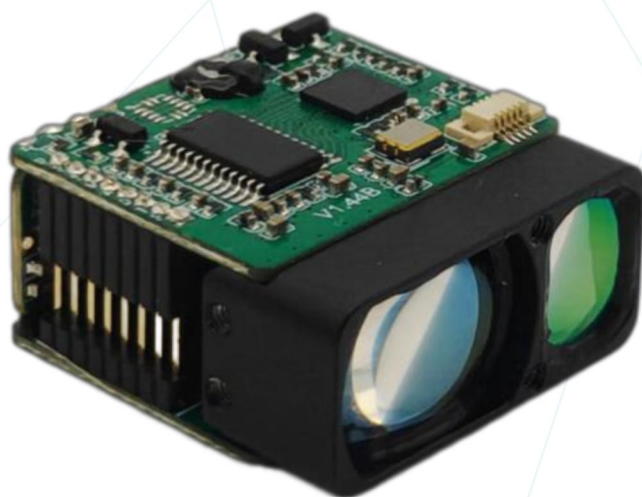




SUNFLASER

OPTOELECTRONIC SOLUTIONS



Product Manual

Product Instruction
Parameter Description

SFA2500BJ

905 nm Miniature Laser Ranging Module

Disclaimer

- This product manual mainly provides the reference specifications and performance parameters of the product. If any specifications are not included in this document, please contact SunFlaser Optics Technology (Chengdu) Co., Ltd. directly for confirmation. Any changes to the contents of this manual or the specific specifications of the ranging module will not be notified separately.
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Safety Instructions

- The laser ranging module should not be viewed directly by the human eye. If observation is necessary, appropriate professional protective equipment must be used. Direct exposure to the laser beam is strictly prohibited to prevent potential eye injury.
- Avoid ranging on targets within the blind zone, especially highly reflective targets such as glass or polished metal surfaces. Avoid operating multiple ranging modules facing each other at close distances. Prevent high-energy laser sources from directly irradiating the receiving optics of the module. During assembly and debugging, keep the receiving lens covered at all times; otherwise, excessive echo signals may cause permanent damage to the detector.
- Pay attention to electrostatic discharge (ESD) protection: the electronic components of the laser ranging module are sensitive to electrostatic discharge. Do not touch any electronic components without proper protective measures.
- Do not touch the optical lenses directly with fingers or hard objects. Avoid using cleaning agents unless necessary, as they may cause irreversible damage to the lens coating and affect product performance.
- Do not use the product under direct sunlight. Do not store the product in highly polluted environments or outside the specified storage temperature range. Avoid exposing the laser ranging module to strong mechanical shocks, such as vibration, impact, or dropping. Avoid placing the module in environments with rapid temperature changes to prevent unpredictable effects on module performance.
- The laser ranging module adopts a certain level of sealing design. However, it is recommended to operate the product in environments with humidity below 80% and maintain a clean operating environment to extend the service life of the product.
- Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not attempt to tamper with or adjust the performance of the module.
- Operate the laser ranging module strictly according to the instructions provided in this datasheet. Avoid sending any operating commands that are not specified in this document.

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01 Product Overview

SFA2500BJ laser ranging module is a new lightweight and compact ranging module operating at a 905 nm wavelength. It provides a maximum ranging distance of ≥ 2500 m and adopts a UART-TTL interface. Equipped with dedicated test software, it facilitates further development by users. The module features a compact size, lightweight design, and reliable performance, making it suitable for applications in aviation, communications, geological exploration, law enforcement, outdoor sports, and other fields.

02 Main Functions

Long-Distance Ranging

The maximum ranging distance reaches up to 2500 m, while supporting short-range measurement for applications including long-distance surveying and close-range detection.

High-Precision Measurement

Measurement accuracy: $\pm(0.7 + 0.004 \times d)$ m when $d > 100$ m, with stable measurement frequency and an error rate of $\leq 1\%$.

Flexible Measurement Modes

Supports single ranging and continuous ranging modes with customizable measurement intervals to meet various testing requirements.

Low Power Consumption Control

Features a low-power mode with standby power consumption ≤ 1 mW. Power states can be switched through the enable pin during normal operation.

Standard Serial Communication

Adopts a UART-TTL (3.3 V) interface with a default baud rate of 115200 bps. Multiple baud rate configurations are supported for easy embedded system integration and development.

Development Support

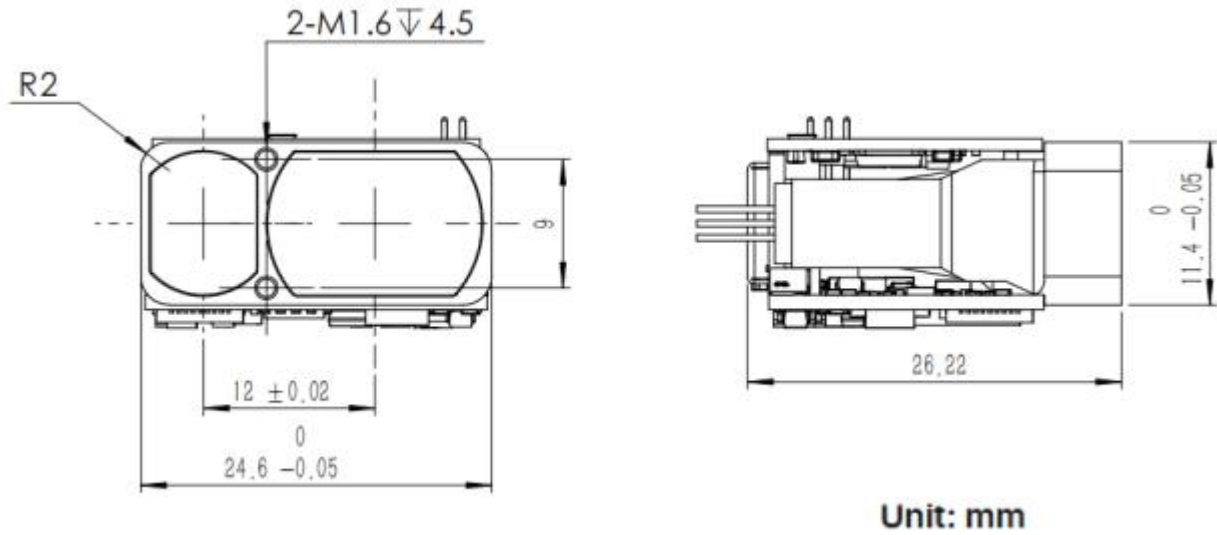
Provides dedicated test software and complete communication protocols, supporting power-on self-test and continuous LD illumination mode debugging to simplify development and integration.

Safety Compliance

The laser safety class is Class 1 (IEC 60825-1), ensuring eye-safe operation and suitability for integration into various civilian applications.

03 Mechanical Dimensions

Mechanical Interface and Center of Gravity Position:



04 Product Performance

Parameter Name		Parameter Value
Electrical Specifications	Maximum Ranging Distance (Sunny day, visibility ≥ 2 km) (Cloudy day, visibility ≥ 10 km)	≥ 2500 m @70% reflectivity, building target. ≥ 3000 m @70% reflectivity, building target.
	Minimum Ranging Distance	5 m @90% reflectivity, white target.
	Ranging Accuracy (Building Target)	$d \leq 100$ m: ± 0.7 m $d > 100$ m: $\pm (0.7 + 0.004 \times d)$ m (where d represents the measured distance, unit: m)
	Measurement Frequency	Normal mode: 3–10 Hz dynamic ranging Long-range mode: 1–10 Hz dynamic ranging
	Measurement Success Rate	$\geq 98\%$
	False Alarm Rate	$\leq 1\%$
	Baud Rate	15200 bps (Factory Default) / 9600 bps
	Interface Type	UART (TTL_3.3V)
	Supply Voltage	DC 3.3 V–5 V (Ripple voltage < 0.1 V)
	Standby Power Consumption	≤ 350 mW (Power ON pulled low) ≤ 1 mW (Power ON pulled high)
	Operating Power Consumption (+3.3 V input)	≤ 1.8 W @25°C ≤ 2.1 W @70°C ≤ 1.6 W @-20°C
	Response Time (System Startup + First Ranging)	Minimum: ≤ 550 ms + 46 ms (Short range) Maximum: ≤ 550 ms + 458 ms (Long range)

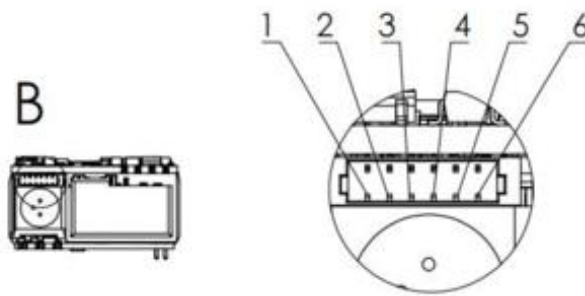
Optical Specifications	Laser Wavelength	905±10 nm, Class 1 (IEC 60825-1)
	Laser Beam Divergence Angle (Typical)	≤10 mrad
	Average Laser Power	7.63 mW
	Receiver Field of View	≈8 mrad
Physical Characteristics	Weight	12±0.5 g
	Dimensions	13.5 mm × 25.1 mm × 27.8 mm
	Shock	1000 g/ms (10 times/s, along the optical axis direction)
	Vibration	5–50–5 Hz, 1 octave/min, 2.5 g
Environmental Adaptability	Operating Temperature	-20°C to +60°C
	Storage Temperature	-25°C to +75°C
	Reliability	MTBF ≥1500 h
	Protection Rating	IP67

05 Electrical Interface

Interface Type: UART (TTL_3.3V)

Baud Rate: 115200 bps (Factory Default)

Connector Model: A1002WR-S-6P



Interface Definition:

Pin	Definition	Description
Pin 1	GND	Ground
Pin 2	Power Supply	3.3–5.5 V DC power supply
Pin 3	NC	No Connection
Pin 4	TTL_TXD	UART transmission output, TTL_3.3 V level
Pin 5	TTL_RXD	UART reception input, TTL_3.3 V level
Pin 6	Enable Pin	Low level enables power-on

Low Power Mode:

In this mode, the device operates with extremely low power consumption. The MCU is powered off and does not respond to any commands.

When ranging is required, pull the enable pin low to enter normal operating mode.

After measurement is completed, pull the enable pin high to enter low-power mode, with power consumption below 1 mW.

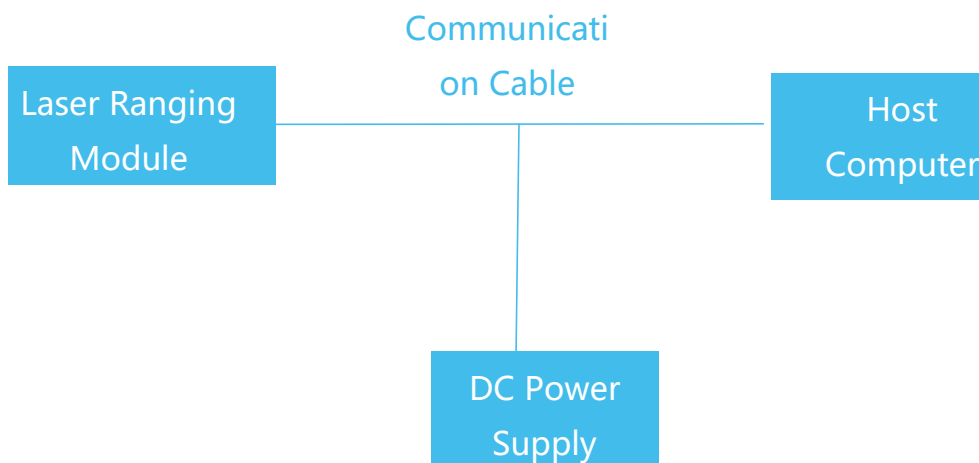
06 Communication Interface

- Communication Protocol: Refer to Appendix 1.

07 Functions and Operation

1. Power-On Operation

- Before powering on, connect the laser ranging module, debugging cable, DC power supply, and host computer according to the electrical interface requirements as shown in the figure below (pay attention to the connector orientation).
- The module can be powered on by connecting the power supply.



Connection Diagram

2. Power-Off Operation

- Before powering off, ensure that all operating processes and tasks have been completed (after the single ranging result has been returned or continuous ranging has been stopped).
- After confirmation, disconnect the power supply to complete the normal shutdown process.

3. Function Description

Ranging Mode Operation:

- Send the "Single Ranging" command to the laser ranging module. The module performs a single distance measurement and reports the ranging status and distance value.

- Send the "Continuous Ranging" command to the laser ranging module. The module continuously performs distance measurements and reports the ranging status and distance values.
- Send the "Stop Ranging" command to stop the ranging operation.

08 Optical Window Selection

1. Material Recommendation

The recommended optical window material is H-K9L optical glass from Chengdu Guangming Optical Glass. H-K9L is one of the most commonly used colorless optical glasses, suitable for laser wavelengths ranging from 300 nm to 2100 nm. It features excellent physical properties and high cost-effectiveness.

2. Processing Recommendations

The wedge angle tolerance of the optical window should be minimized. A wedge angle tolerance of $\leq 3'$ is recommended (tolerance grade $\leq$ Grade 7).

The optical surfaces of the window should be as smooth as possible. A recommended arithmetic average roughness (Ra) value is 0.012.

3. Coating Recommendations

For the 905 nm laser ranging module, it is recommended that the optical window be coated with an anti-reflection coating for the wavelength range of 855 nm to 955 nm, with a transmittance of $\geq 99.5\%$. Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other specifications shall comply with GJB2485-95, with a transmittance of $\geq 98\%$.

4. Optical Window Dimensions and Usage Recommendations

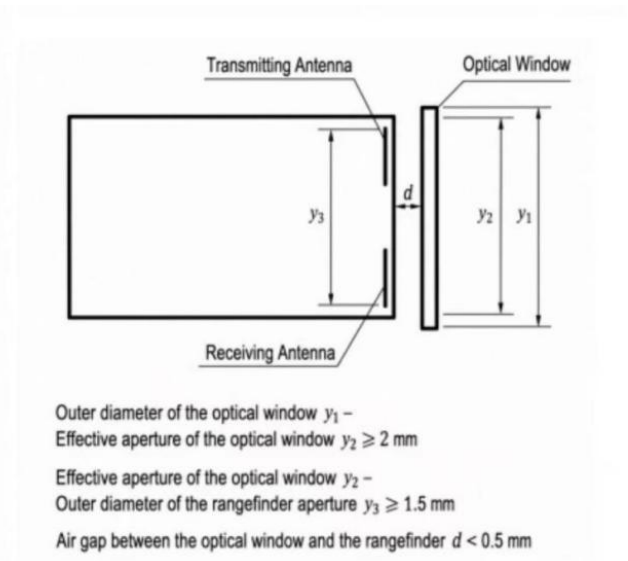
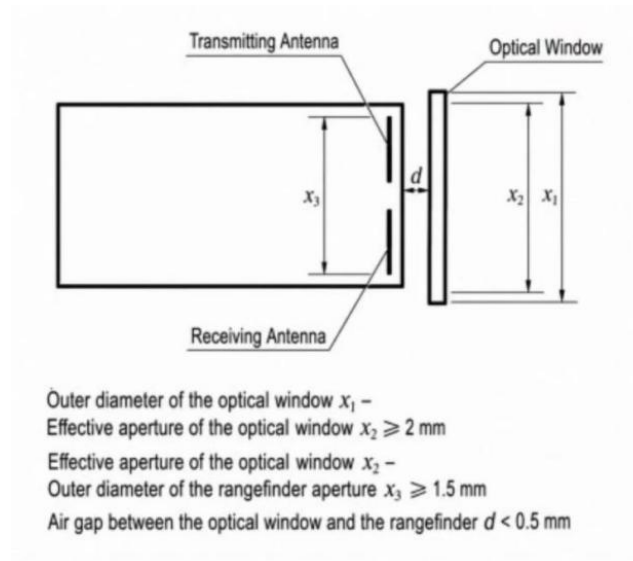
The effective aperture of the optical window depends on the specific product design. The window dimensions should ensure that the difference between the outer diameter of the optical window and its effective aperture is ≥ 2 mm, and the difference between the effective aperture and the outer diameter of the ranging module aperture is ≥ 1.5 mm. The schematic diagram is shown below.

Since the optical window absorbs a certain amount of laser energy, the window thickness is recommended to be controlled within 2–4 mm depending on its dimensions.

Due to the high transmittance of the optical window, it is recommended that the transmitting optical axis be parallel to the normal axis of the window. Meanwhile, the air gap between the optical window and the ranging module should be less than 0.5 mm. The positional relationship between the optical window and the two lens barrels is shown in the figure below.

During use, the optical window must be kept clean and transparent. The optical surface should be free from stains, fingerprints, dust, and other contaminants.

Optical Window Dimensions and Placement Diagram



09 Maintenance Instructions

1. Packing List

The items supplied with the product are listed in the table below. (Optional accessories may not be included in the list. Please refer to the actual delivered items.)

No.	Item	Quantity	Remarks
1	Laser Ranging Module	1 unit	
2	Certificate of Conformity	1 copy	
3	Packaging Box	1 pcs	
4	Power Cable	1 pcs	

2. Cleaning

2.1 Cleaning of Optical Components

- Dust particles should be removed using an air blower.
- Fingerprints should first be removed by gently wiping with degreased cotton dipped in a small amount of alcohol-ether mixture, followed by cleaning with a clean lens cleaning cloth.

2.2 Cleaning of Mechanical and Electronic Components

- Clean mechanical and electronic components with alcohol while the device is powered off, and allow them to air dry completely before use.
- Keep the ranging module, connector plug, and cables away from moisture and contaminants whenever possible.
- Ensure the equipment is completely dry before packaging.

3. Inspection and Maintenance

3.1 General Inspection

Visual inspection and power-on inspection shall be performed before the initial use of the product and after replacement of functional modules. For products in normal operation, only power-on inspection is required before use.

The visual inspection steps are as follows:

- Check whether the product appearance is normal.
- Check whether the cable connection is correct and ensure all connections are secure.
- The power-on inspection steps are as follows:
 - Perform the power-on procedure and check whether the standby current is within the specified range.
 - Start the self-test function to verify product operation.
 - Perform the shutdown procedure after completing the inspection.

3.2 Periodic Maintenance

The laser ranging module does not require regular maintenance under normal operating conditions. Maintenance is required if the product has been stored in a dust-free environment for more than one year. Maintenance mainly includes general inspection and power-on inspection.

General inspection shall be performed with the product powered off. The procedures are as follows:

- All markings and numbers on the product and test cable connectors (plugs and sockets) shall be correct and clearly visible.
- All screws on the panel shall be properly tightened.
- Ensure that the optical glass is free from visible defects or contaminants that may affect normal observation, including spots, pits, water stains, mold, fingerprints, dust particles, cracks, or other attachments.

A complete inspection and maintenance procedure shall be performed with the laser ranging module powered on, including:

- Turn on the product power supply.
- Perform the power-on procedure and verify that the standby current is within the specified range.
- Start the product self-test function and complete the self-test procedure.
- Complete the shutdown procedure.

4. Fault Analysis and Troubleshooting

The following section only lists possible fault phenomena that may be caused by factors other than the product itself. If any other technical issues occur during product use, please contact our company.

Measured distance is shorter than the actual distance	When testing under severe haze, heavy humidity, fog, or water mist conditions, water droplets or particles in the air may reflect the laser signal and be detected as targets, resulting in shorter measured distances. It is recommended to perform measurements under suitable environmental conditions.
Reduced ranging performance, unable to detect targets at the theoretical maximum range	①Low target reflectivity reduces the returned laser energy, which may affect ranging performance. When the target is a low-reflectivity object such as a lake surface, wet road, or glass, the actual ranging performance may not reach the theoretical maximum range.

	②If the laser ranging module does not point vertically toward the target and forms an angle with the target surface, the returned laser energy will decrease under oblique measurement conditions, resulting in reduced ranging performance. ③Since the laser wavelength overlaps with part of the solar spectrum, directly pointing the ranging module toward sunlight may reduce ranging performance.
No response when using the company's complete communication cable to connect with the PC	Check whether the communication cable is damaged or broken (the 6-pin communication cable is relatively thin and may be prone to breakage).
No response when using a self-prepared communication cable	Check whether the external wiring sequence of the module is correct. Confirm that the communication cable has no short circuit, open circuit, or poor contact, and verify that the serial communication wires are not connected in reverse.

5. Packaging, Transportation and Storage Requirements

5.1 Packaging

After unpacking, if the product needs to be stored again, it shall be repackaged using the original packaging. When the product needs to be returned to the manufacturer, the original packaging should be used whenever possible. If other packaging methods are adopted, they shall not cause any performance degradation or damage to the product.

5.2 Transportation

Repackaged products may be transported by road, railway, air, or sea. During transportation, the packaged product shall be securely fixed to the transport vehicle to prevent impact, rough handling, rain, snow, or other adverse conditions.

10 After-Sales Service

If a product failure occurs and the complete unit needs to be returned to the manufacturer for fault analysis, diagnosis, and repair, SunFlaser provides a one-year warranty service and lifetime technical support from the date of product delivery. During the warranty period, SunFlaser will provide free replacement or repair for problems caused by product quality issues. For failures caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and protect the product carefully during use. If you have any questions regarding operation, maintenance, or service, please contact our after-sales support team at any time.

11 Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1

1. Test Guide

Operation Steps

Step 1: Insert the data cable into the ranging module to provide power supply and output measurement data simultaneously.

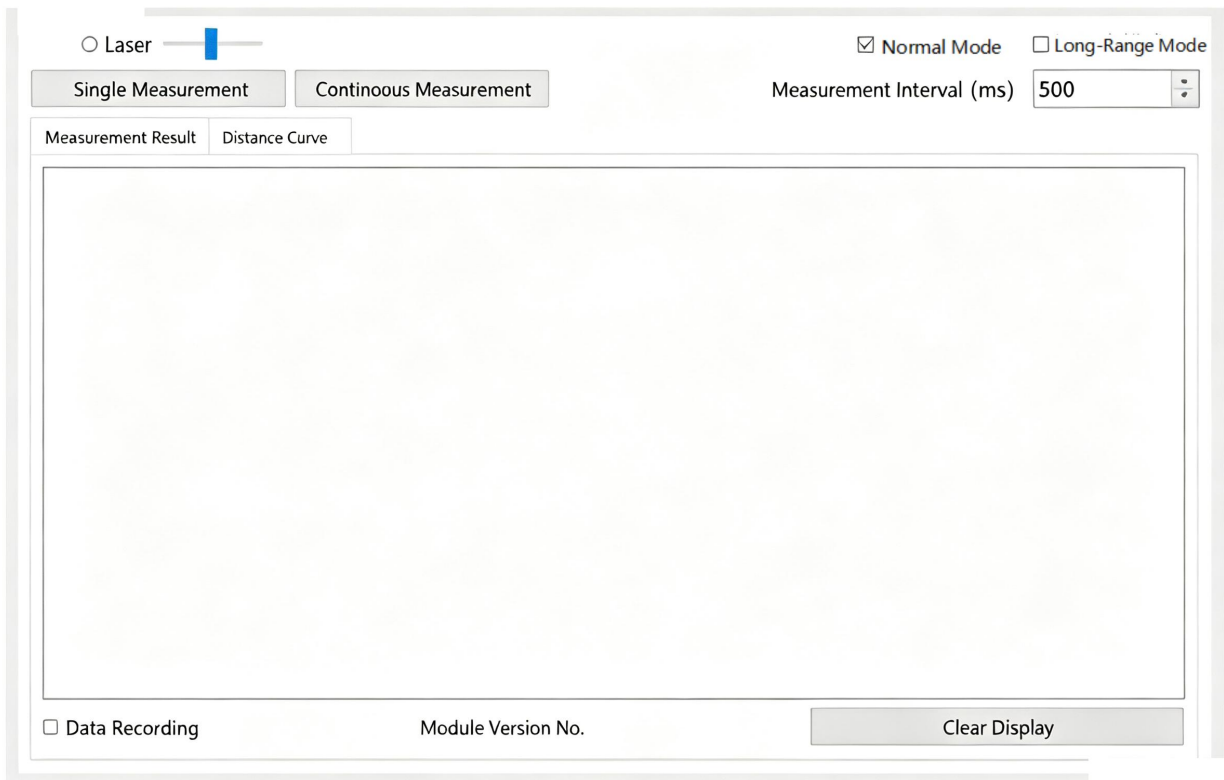
Note 1: Use the original factory-supplied cable and strictly maintain the power supply voltage within the range of 3.3–5 V.

Note 2: The enable pin must be pulled low (connected to GND).

Step 2: Install the software and use a CH430 or CP2102 module to convert the serial signal to USB for connection to the computer.

Note 1: Connect the module TXD to the TTL-USB module RXD, and connect the module RXD to the TTL-USB module TXD.

Step 3: After the software installation is completed, open the display interface.



Step 4: Debugging and Testing

1. Select the COM Port: Select and configure the corresponding COM port in the software according to the computer's serial port number.
2. Baud Rate Setting: Open the software interface to configure the baud rate. Available baud rates include 9600 bps, 115200 bps, and others (refer to the Communication Protocol).

2. Protocol Format

Communication Method: Serial communication is adopted.

Baud Rate: 115200 bps (Default).

Data Bits: 8 bits.

Frame Length: 8 bytes.

Data Protocol									
		Frame Header H	Frame Header L	Function Code	D1	D2	D3	D4	Checksum
Transmission		55	AA						SUM (Function Code + DATA1 + ... + DATA4)
Response		55	AA						SUM (Frame Header H + Frame Header L + ... + DATA4)

Measurement Commands									
Single Ranging	Command	55	AA	88	FF	FF	FF	FF	SUM[3: 7]
		55 AA 88 FF FF FF FF 84							
	Response	55	AA	88	STA	FF	DIS_H	DIS_L	SUM[1: 7]
		STA = Other values: Measurement failed; STA = 1: Measurement successful. DIS_H: High byte of the measured distance value; DIS_L: Low byte of the measured distance value. The returned data is transmitted in hexadecimal format. The actual measurement value is multiplied by 10 before output.							
Continuous Ranging	Command	55	AA	89	FF	FF	FF	FF	SUM[3: 7]
		55 AA 89 FF FF FF FF 85							
	Response	55	AA	89	STA	FF	DIS_H	DIS_L	SUM[1: 7]
		STA = Other values: Measurement failed; STA = 1: Measurement successful. DIS_H: High byte of the measured distance value; DIS_L: Low byte of the measured distance value. The returned data is transmitted in hexadecimal format. The actual measurement value is multiplied by 10 before output.							
Stop Ranging	Command	55	AA	8E	FF	FF	FF	FF	SUM[3: 7]
		55 AA 8E FF FF FF FF 8A							
	Response	55	AA	8E	STA	FF	FF	FF	SUM[1: 7]
		STA = 0: Continuous ranging stop failed; STA = 1: Continuous ranging stopped successfully.							
Angle Measurement	Command	55	AA	8A	FF	FF	FF	FF	SUM[3: 7]
		55 AA 8A FF FF FF FF 86							
	Response	55	AA	8A	STA	FF	ANG_H	ANG_L	SUM[1: 7]
		STA = 0: Measurement failed; STA = 1: Measurement successful. ANG_H: High byte of the measured angle value; ANG_L: Low byte of the measured angle value. The returned data is transmitted in hexadecimal format. The actual measurement value is multiplied by 10 before output. The angle measurement function is only available for modules equipped with an angle sensor.							

Power-On Self-Test (POST)									
Self-Test Information	Response	55	AA	80	STA	00	00	ErrCode	SUM[1: 7]
			STA = 0: Power-on initialization failed, and ErrCode indicates the error code. STA = 1: Power-on initialization completed successfully.						

System Settings									
Baud Rate	Command	55	AA	TYPE	FF	FF	FF	FF	SUM[3: 7]
		TYPE = 01: Set baud rate to 9600 bps TYPE = 02: Set baud rate to 14400 bps TYPE = 03: Set baud rate to 19200 bps TYPE = 04: Set baud rate to 38400 bps TYPE = 05: Set baud rate to 56000 bps TYPE = 06: Set baud rate to 57600 bps TYPE = 07: Set baud rate to 115200 bps TYPE = 08: Set baud rate to 128000 bps TYPE = 09: Set baud rate to 230400 bps Note: The baud rate setting will not take effect immediately. The device must be restarted for the new setting to become effective.							
	Response	55	AA	TYPE	STA	FF	FF	FF	SUM[1: 7]
		STA = 0: Setting failed. STA = 1: Setting successful.							
LD Continuous Illumination Mode	Command	55	AA	86	FF	FF	FF	STA	SUM[3: 7]
		STA Definition: STA = 0: Disable LD continuous illumination mode STA = 1: Enable LD continuous illumination mode Command to Enable LD Continuous Illumination: 55 AA 86 FF FF FF 01 84							

		Command to Disable LD Continuous Illumination: 55 AA 86 FF FF FF 00 83
	Response	55 STA = 0: Enabled STA = 1: Disabled STA = FF: Operation failed Note: Do not keep the LD continuous illumination mode enabled for a long time.

1. The checksum calculation method is different between transmitted and received data. Please distinguish them carefully.
2. Checksum Calculation: The checksum is the lower 8 bits of the sum of all bytes to be verified.
3. All data is transmitted and received in hexadecimal format.